

The next delivery for consideration is 8, this can be allocated to route 2, as the bicycle will arrive at 14:01.

The next customer for consideration is customer 1, as they have a 09:00 to 10:00 window they cannot be served on route 2, so another route is started using a van. This continues until the genome has been fully decoded.

Routes			
1V		2B	
5	13:00	2	09:00
		4	13:00
		D	
		8	14:01

Implementation

The multi-modal home deliveries example has been implemented in Java; the source code may be found within the repository that accompanies this book. An outline class diagram is shown in Fig. 6.1.

In order to implement MAP-Elites, we need an efficient implementation of the multi-dimensional archive. Our implementation is contained within the class Archive. The efficiency of this data structure is crucial to the viability of the algorithm, an inefficient structure will result in an algorithm that is too slow to be of practical use.

The capacity of the archive is

$$c = b^d$$

Where

c is the capacity needed

b is the size of the scaled range per dimension

d is the number of solution characteristics (dimensions).

In the FabFoods problem, which has a scaled range of 10 over each of the 9 dimensions, the archive structure needs to have a capacity of 10^9 . Attempting to create an array of that length in Java will result in the Java Virtual Machine running out of heap space (although it may be possible to overcome this on some systems by using `-Xmx` option to increase heap space). The storage mechanism adopted here for use within archive is a map which allows items to be added and the tree expanded as required.

The Multi-dimensional Archive

Solutions within the archive are identified by a *key* which comprises an array of integers, one value for each dimension, each item specifying the scaled value of the